

Aaron Yangelo

SENIOR SOFTWARE ENGINEER - AGENTIC ENGINEERING & PRODUCT DELIVERY

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AI-native engineer and product builder with 10 years across defense and public safety, combining rigorous systems design, hands-on technical leadership, and agentic workflows to ship reliable software and measurable operational outcomes.

Experience

Flock Safety

Remote

Senior Software Engineer - Sharing, Trust, and Compliance

March 2026 - Present

- Architected and implemented a customer-controlled search policy system that lets agencies restrict which criminal offense categories external organizations may use when searching their data; authored the system design and aligned Product, Design, and Engineering partners across four teams, driving approximately **\$500K ARR in its first month**. *TypeScript, Node.js, PostgreSQL, Redis*
- Introduced agentic coding to the Sharing, Trust & Compliance team through repeatable planning, implementation, parallel-review, and validation workflows, cutting average Jira work-item completion time by **43%** and increasing support-ticket throughput by **66.3%** despite **35.9% higher volume**.
- Added the policy-enforcement service to the product's highest-volume, most critical search workflow, which handles more than **100K searches/day**, and reduced its p95 latency from **3–5 seconds to under 200 ms**, introducing real-time enforcement with no perceptible increase in end-user search latency.
- Resolved release-blocking performance problems in the redesigned Sharing experience across frontend and backend, improving six measured p95 stages by **15%–71%** in pre-release validation and unblocking its planned customer rollout.
- Launched and continue to publish a biweekly AI Enablement Digest that translates internal use cases, reusable engineering skills, practical exercises, industry developments, and AI Developer Enablement discussions into actionable guidance; companywide AI adoption increased **4x** after launch.

Innovative Defense Technologies

Mt. Laurel, New Jersey

Senior Software Engineer - Data and Analytics Platform

May 2016 - March 2026

- Owned architecture and roadmap for Navy's Ship Self Defense System (SSDS) analysis platform, leading a 5-person full-stack team through modernization, scaling, and sustainment of mission-critical reliability and throughput across multiple programs.
- Acted as Release Lead for the Aegis analytics platform, coordinating tasks for 14 engineers and delivering 11 major releases with **100% on-time delivery**. Modernized verification frameworks and mentored engineers to scale the release process independently. *JIRA*
- Led design and end-to-end implementation of the SSDS analytics platform, architecting a high-throughput data pipeline ingesting **terabytes of data per run**. Enabled mission operators to validate fleet software updates **7x faster** than manual baselines, accelerating readiness and decision cycles. Optimized SQL and decoding logic to achieve an additional **5.2x performance gain**. *C++, Python, Java, PostgreSQL, TypeScript, Docker, REST APIs, DOORS*
- Built an internal analytics and visualization suite that standardized **data observability** practices across programs, enabling cross-comparison of experimental test runs, runtime metrics, and regression trends. Reduced detection-to-fix time by **8-15x**, establishing the benchmark for future analysis workflows. *Java, PostgreSQL*
- Developed a **dependency-free testing framework** replacing full database dependencies with CSV-based mocks, reducing validation runtimes **12x** and standardizing reproducible regression testing across the data engineering org. *C++, Catch2*
- Architected virtualization of a legacy multi-service tactical system into a reproducible, cloud-based development platform, eliminating hardware lab dependencies and enabling distributed engineering teams and mission operators to build, test, and validate software anywhere. Achieved **orders-of-magnitude faster iteration** while maintaining latency within **0.1% of physical baselines**. *vSphere, Nexus, Bash, TCL/TK, MS Azure, DOORS*

Freelance Software Engineer

Contracted on Per Project Basis

2018-Present

- **Liquid Trucking:** Architected and delivered an enterprise integration layer connecting Trimble fleet maintenance systems with BlackBerry Radar asset tracking, automating metric collection, reporting, and maintenance scheduling. Enabled data-driven fleet decisions and produced estimated annual savings of **\$1.43M** through reduced downtime and manual overhead. *Python, REST APIs, JWT Auth*
- **PlanetBravo (K-12 Technology Education):** Designed and deployed a Google Sheets automation platform for lesson planning, weekly dashboards, and multi-campus oversight—**cutting review time from 6 hours to 1.5 hours** and **improving administrative accuracy 4x**. Provided ongoing advisory support to align product workflows with educational data privacy standards. *JavaScript, Google Sheets API*
- **Ciocca Automotive:** Led full-stack development of a customer portal automating position tracking and communication for Corvette Z06 order waitlists, replacing manual lookup processes with real-time visibility. System scaled to **thousands of concurrent users** and became the dealership group's standard for customer engagement. *Node.js, HTML/CSS, JavaScript, Google Sheets API*

Education

M.S. Computer Science

Georgia Institute of Technology | Computing Systems

B.S. Electrical and Computer Engineering

Rowan University | Minor: CS, Systems Engineering

Technical Skills

Languages: Python | Java | C/C++ | TypeScript/JavaScript |

SQL (PostgreSQL, MySQL, Clickhouse) | Bash

Frameworks & Libraries: React | Node.js | TailwindCSS | FastAPI | Express | Playwright | Pandas | NumPy | Test Driven Development

Architecture & Infrastructure: Microservices | Event-Driven Systems | Distributed Systems | REST APIs | Docker | Kubernetes | Redis | AWS, Azure

Security & Data: Data Privacy & Governance | Policy Engines | Secure API Design | Grafana | Data Dog | Prometheus

Agentic Development: Claude Code | Codex | Local LLMs | Multi-Agent Orchestration | Multi-Agent Review | Context Handoffs | Loops | Skill Development

Select Projects

Full project list available at AaronYangelo.com

Malpais CMMS: A cloud-based fleet maintenance enterprise platform for multi-shop commercial vehicle operations, unifying work orders, preventive maintenance, parts inventory, labor, compliance, and accounting in role-based desktop and mobile workflows. [Watch Demo Video](#)

Agent Orchestration, React, TypeScript, Supabase, PostgreSQL, Tailwind CSS

Projection Show Engine: A local-first projection-mapping engine with native GPU rendering and a responsive web control room for live surface calibration, synchronized video, lighting and audio timelines, media management, and atomic show builds. [Watch Demo Video](#) | [See on GitHub](#)

Agentic Programming, Python, FastAPI, ModernGL, React, TypeScript, GStreamer